

3. We let $x = m\alpha$. Then,

$$\lfloor \frac{\lfloor m\alpha \rfloor n}{\alpha} \rfloor = \lfloor \frac{\lfloor x \rfloor n}{\frac{x}{m}} \rfloor = \lfloor \frac{\lfloor x \rfloor mn}{x} \rfloor$$

.